Applied Linear Algebra

Linear Functions

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Functions

We write $f : \mathbb{R}^n \to \mathbb{R}$ to mean that the function f maps real n-vectors to real numbers.

Function f is a scalar-valued function of n-vectors.

Example

$$f(\mathbf{x} : \mathbb{R}^n) = \mathbf{1}_n^{\mathrm{T}} \mathbf{x}$$

Linear Functions

A linear function satisfies the following two properties:

- *Additivity.* $f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$ for all n-vectors $\mathbf{x}$ and $\mathbf{y}$.
- *Homogeneity.* $f(\alpha\mathbf{x}) = \alpha f(\mathbf{x})$ for all n-vectors $\mathbf{x}$ and all scalars α.

Prove the following

Definition

A function f is *linear* if it satisfies the *superposition principle*:

$$f(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha f(\mathbf{x}) + \beta f(\mathbf{y})$$

for all n-vectors $\mathbf{x}$ and $\mathbf{y}$, and all scalars α and β.

Prove the following

Definition

A function f is *linear* if it satisfies the *superposition principle*:

$$f(\alpha\,\mathbf{x} + \beta\,\mathbf{y}) = \alpha\, f(\mathbf{x}) + \beta\, f(\mathbf{y})$$

for all n-vectors $\mathbf{x}$ and $\mathbf{y}$, and all scalars α and β.

Hint

- Substitute $\alpha\mathbf{x}$ with $\mathbf{u}$ and $\beta\mathbf{y}$ with $\mathbf{v}$.
- Apply *addivity* and *homegeneity* to arrive at the result.

Vector Inner Product Function

Consider the vector inner product function

$$f(\mathbf{x}) : \mathbb{R} = \mathbf{a}^{\mathrm{T}}\mathbf{x} = a_1 x_1 + \cdots + a_n x_n$$

where $\mathbf{a}$ is a fixed n-vector, and $\mathbf{x}$ is a variable n-vector.

The function f is called a *linear function* of $\mathbf{x}$.

The vector $\mathbf{a}$ is called the *coefficient vector* of the linear function f.

$\mathbf{a}$ is also called the *weight vector* of the linear function f.

Prove the following

The vector inner product function

$$f(\mathbf{x}) : \mathbb{R} = \mathbf{a}^{\mathrm{T}}\mathbf{x} = a_1 x_1 + \cdots + a_n x_n, \quad \mathbf{a} : \mathbb{R}^n, \ \ \mathbf{x} : \mathbb{R}^n$$

is linear.

Vector Inner Product Function

The vector inner product function

$$f(\mathbf{x}) : \mathbb{R} = \mathbf{a}^{\mathrm{T}}\mathbf{x} = a_1 x_1 + \cdots + a_n x_n, \quad \mathbf{a} : \mathbb{R}^n, \ \ \mathbf{x} : \mathbb{R}^n$$

is linear.

Proof

Given the inner product function $f(\mathbf{x}) = \mathbf{a}^{\mathrm{T}}\mathbf{x}$, we can verify that it is linear:

$$\begin{aligned} f(\alpha\,\mathbf{x} + \beta\,\mathbf{y}) &= \mathbf{a}^{\mathrm{T}}(\alpha\,\mathbf{x} + \beta\,\mathbf{y}) \\ &= \mathbf{a}^{\mathrm{T}}(\alpha\,\mathbf{x}) + \mathbf{a}^{\mathrm{T}}(\beta\,\mathbf{y}) \\ &= \alpha(\mathbf{a}^{\mathrm{T}}\,\mathbf{x}) + \beta(\mathbf{a}^{\mathrm{T}}\,\mathbf{y}) \\ &= \alpha f(\mathbf{x}) + \beta f(\mathbf{y}) \end{aligned}$$

Linear Combination of Vectors

Definition
The linear combination of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ with scalars $c_1, c_2, \dots, c_n$ is the vector

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \sum_{i=1}^{n} c_i\mathbf{v}_i$$

The scalars are called the *weights* of the linear combination.

By definition, a linear combination involves only finitely many vectors.

Linear Independece and Span

Definition (Linear Independence)

A set of vectors where no single vector can be written as a linear combination of the others.

Definition (Span)

The complete set of all possible vectors one can reach by changing the scalar values in a linear combination.

Definition

If a plane P contains two points A and B, then it also contains the line $\overrightarrow{AB}$.

This expresses the intuitive notion of “flatness”.

Non-planar surfaces do not satisfy this property.

Definition (Geometric Linearity)

A subset S of $\mathbb{R}^n$ is geometrically linear if S contains all of the lines through the points of S.

Algebraic Linearity

Definition
A a multivariable polynomial function is called algebraically linear if every term has degree 1.

This is a “strict” definition. We may also include constants, i.e., terms of degree 0, in which it becomes an *affine* function.

Affine Functions

Definition (Affine Function)

An *affine function* is a linear function followed by a translation. It can be written as

$$f(\mathbf{v}) = \mathbf{L}(\mathbf{v}) + \mathbf{b}$$

where

- $\mathbf{L}$ is a linear function, and
- $\mathbf{b}$ is a fixed translation vector.

Affine Function - An Example

Given the general definition of an affine function

$$f(\mathbf{v}) = \mathbf{L}(\mathbf{v}) + \mathbf{b},$$

and the values of the variables,

$$\mathbf{L} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}.$$

caculate the value of the function f.

Affine Combination

A *linear combination* has the form

$$\sum_{i=1}^{k} c_i \mathbf{v}_i,$$

with no restriction on the coefficients.

An *affine combination* has the additional constraint

$$\sum_{i=1}^{k} c_i = 1.$$

The coefficients may be positive, zero, or negative.

Convex Combinations

In an *affine combination* $\sum_{i=1}^{k} \mathrm{c}_i = 1$, the coefficients may be positive, zero, or negative.

In addition, if

$$\mathrm{c}_i \geq 0 \qquad \text{for all } i,$$

then the affine combination is called a *convex combination.*

Regression Model

An n-vector $\mathbf{x}$ is a *feature vector* of n *features.*

A *regression model* is the affine function of $\mathbf{x}$ given by

$$\hat{\mathrm{y}} = \mathbf{x}^{\mathrm{T}}\beta + \mathbf{v}$$

where β is an n-vector and $\mathbf{v}$ is an n-vector.

The entries of $\mathbf{x}$ are also called the **regressors** or **predictors**.

regressors literally mean "the things that cause the *prediction* or *regression*."

$\hat{\mathrm{y}}$ is called the *prediction*.

Applied Linear Regression

Prediction of Intensive Care Unit Length of Stay (LOS) in the *MIMIC-IV Dataset*
"... four different classification and regression models for predicting LOS in the ICU are utilized... The first is the common logistic/linear regression model, which incorporates a linear combination of all features in modeling..."

- body temperature
- chloride
- creatinine
- heart rate
- mean corpuscular volume (MCV)
- O2 saturation, and
- respiratory rate